

2408.TW - Nanya Technology | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | BOTTLENECK - qualified DRAM supply, with a capacity clock | High Conviction

Analytical Base: 45/60 (75%). Valuation: 6x forward four-quarter IFRS-style operating profit.

Snapshot: TWD 516.00 per share on September 21, 2026 close; 3,450.206 million shares, implying TWD 1,780,306.24 million equity value. Source [N7]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: TWD 1,178.41/share. Q1 2028: TWD 1,276.37/share.

All financial model amounts are TWD millions, shares in millions, unless stated. Calendar and fiscal quarters coincide; fiscal year ends December 31. Q1/Q2 are Actual; July/August revenue is Preliminary; all full quarters from Q3 2026 are Model.

1. Thesis, bottleneck and expanded TAM

Nanya designs and manufactures DRAM for PCs, communications, consumer devices, industrial/automotive uses and a growing server/AI infrastructure base. It sells the volatile working-memory component; it is neither Nanya PCB nor a NAND manufacturer. Framework DB6 is the principal cycle, with DB1 networking and physical-AI adoption widening applications.

Large DRAM suppliers allocating advanced capacity to HBM and premium products reduce supply available to mature-node and specialty customers. Nanya benefits through contract repricing and technology conversion. The immediate earnings mechanism is price on constrained existing output, followed by more bits. HBM industry growth is an indirect supply-allocation tailwind, not proof that Nanya sells leading accelerator HBM.

Qualification, process know-how and a functioning wafer fab are real barriers. They are not a permanent sole-source position: added industry capacity, alternative DRAM suppliers and customer product redesign can remove scarcity. Current pricing is a bottleneck phase of the S-curve; 1C/1D/1E and customized AI memory are earlier technology transitions. Do not treat a peak commodity margin as an irreversible structural floor. [N2,N3]

The 2025 Annual Report Market Status and risk sections identify Samsung, SK hynix, Micron and CXMT alongside Nanya. Its 2025 share was about 1.4%; that stale historical share is not multiplied by a 2030 all-memory TAM. CXMT capacity and product overlap are explicit competitive threats. The named competitor screen establishes no verified overlooked listed sub-\$2B monopoly. Nanya needs reliable qualification and niche customer retention as well as cost reduction. [N6]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
Phase 1: current computing and AI infrastructure	DRAM working memory in servers, eSSD controllers, networking, PCs and devices	100% weight to established demand; do not count an entire eSSD as Nanya NAND revenue.
Phase 2: physical AI, 2026-30	Low-power/custom memory for edge inference, industrial control, robotics and vehicles	50% conceptual confidence; product mix within the DRAM pool, not an added memory market.
Phase 3: ubiquitous compute, 2028+	Customized low-power/high-bandwidth memory for new embedded architectures	20% conceptual confidence; no separately verified dollar TAM or unannounced design-win revenue.

Demand-side Plan B uses Q2 DRAM revenue x qualified bit-volume index x realized-ASP index, calibrated to 82,522.846 with zero anchor error. Customer-specific capex, purchase quantities and wallet shares are not disclosed; multiplying aggregate hyperscaler capex by an invented Nanya share would overstate evidence. Management identifies AI/server demand above 20% of sales and a high-teens 2026 bit-growth goal. Monthly revenue then constrains near-term recognition. The annual report cites approximately 16-20% industry bit-demand growth in 2026-29; model volume growth is tested against that range and disclosed fab timing, rather than assumed to match it exactly. [N1,N2,N4,N6]

2. Latest evidence and financial anchor

The April 13 and July 10 management presentations were read. They show a shift from price recovery to strong cash generation and new-fab preparation. The linked webcasts were found, but accessible full management Q&A transcripts were not obtained; this is an explicit earnings-protocol evidence gap. Slides are not described as transcripts. No analyst question or unsourced call summary is treated as management guidance. [N2,N3]

July revenue was 43,867.609 and August 44,690.268. The company identifies the July contract-reset effect. September is modeled at August x 1.02 = 45,584.073, producing Q3 revenue 134,141.950. This replaces a generic industry ASP extrapolation where company monthly evidence is stronger. Q3 is still Model because September and full-quarter costs have not been reported. [N4,N5]

Current production and already recognized customized AI products remain in the DRAM anchor. The new fab is a later capacity source, with tool installation in early 2027 and first-phase 30,000 wafer starts/month planned in 2028. Long-term full capacity is 45,000; neither is immediately saleable qualified output. No separate speculative HBM revenue vector is inserted. [N2,N3]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026	Actual	49,086.93	33,316.12	30,111.28	N1 H1 less Q2
Q2 2026	Actual	82,549.07	65,619.21	60,826.28	N1 reviewed statement

Q2 DRAM sales 82,522.846 plus other revenue 26.227 = 82,549.073. Allocate the small residual a 30% proxy gross margin; DRAM GM solves to $(65,619.208 - 26.227 \times 30\%) / 82,522.846 = 79.507\%$. This preserves reported gross profit exactly. Reported net GP 65,619.208 less R&D 3,229.782 and selling/administration 1,563.149 = OP 60,826.277. DDR-generation GP is not separately disclosed, so no fabricated DDR4/DDR5 margin ledger is claimed. The DRAM pool is the available product-level anchor. [N1]

No numeric Q3 revenue or OP guidance was located. Management expects improvement and high-teens 2026 bit growth; the monthly-anchored Q3 model is consistent with the direction. The Q3 realized-ASP index is solved from revenue with flat bits and neutral FX, not asserted as a disclosed contract-price increase. If actual bits or FX differ, this decomposition must be re-estimated while retaining reported revenue. [N2,N4]

Balance sheet and capital allocation

Q2 cash was approximately 216,757, debt 18,400 and net cash 198,400; operating cash flow was 55,013. April customer equity financing raised about 78,718 and materially contributed to liquidity. This is equity, not customer prepayment, so it earns no prepayment score bonus. Approved 2026 capex is capped at 52,000, with H1 spending about 6,900. Full 45,000-wafer new-fab investment is about USD16B over time, not one year of approved capex. Strong liquidity does not eliminate future financing or depreciation risk. Cash is not added again to the framework ceiling. [N2,N3,N6]

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis /20	15	Qualified DRAM manufacturing and constrained supply; substitutable and exposed to major rivals.
Cycle Stage /10	9	Contract repricing already recognized; volume capacity follows later.
TAM Expansion /10	8	A memory primitive serving infrastructure and physical AI, with credible custom/low-power paths.
Catalysts /5	5	Monthly revenue, 2026 contract reset, 2027 tools and 2028 capacity entering forward windows.
Macro /10	9	AI memory allocation supports specialty scarcity; consumer demand remains price-sensitive.
Alpha /5	3	Local monthly and contract-reset evidence improves timing; major industry narrative is well known.
Risks	-4	Competition, execution, cycle/customer demand and capital commitments.
Total	45/60 = 75%	High Conviction

45/60 = 75% is the independently derived Base for this new entry. Taiwan location alone receives no penalty. No governance penalty or prepayment bonus is asserted without specific evidence.

4. Quarterly operating model

One base case with 18 operating quarters and 15 complete forward-four-quarter ceilings. Two available revenue pools: DRAM and the immaterial reported residual. Gross margin is derived from an independent unit-cost path, not set as a fixed operating margin. Ordinary depreciation remains in cost of sales. The result is an IFRS-style OP forecast, not net income or EPS.

Revenue vector	Driver and source	Confidence
DRAM	N1 actual product revenue; N2 bit/price/fab outlook; N4-N5 monthly contract-reset anchor. Volume, ASP and unit-cost indices are analyst assumptions.	MEDIUM near-term revenue / LOW future margins and 2028+
Other revenue	N1 reported residual; no detailed margin disclosure. 30% GM and modest growth are immaterial proxies.	LOW

Q2 2026 = 1 for bit volume, ASP and unit cost. DRAM revenue = 82,522.846 x volume x ASP. DRAM GM = 1 - (1 - 79.507% anchor GM) x unit-cost index / ASP index. Q3 volume is flat; Q4 +2%; 2027 rises from 1.06 to 1.18. After Q3 calibration, ASP increases 10% in Q4, 10%/5%/3%/0% through 2027. These are explicit company-specific analyst extrapolations of scarcity, not TrendForce DRAM guidance. The user NAND +23%/+20%/+15% schedule is not applied. The user-provided BofA image informs a 5% cumulative DRAM ASP reduction during 2028; 2029/30 recover 3%/2% cumulatively. Unit cost rises to 1.16 by Q4 2027 and 1.25 in Q1 2028 for ramp/depreciation pressure, then eases with yield improvement. R&D is 3.5% of sales through 2027, 3.7% in 2028, then 3.6%/3.5%; SG&A stays 1.8%. These ratios create higher absolute spending as sales expand. Their values are modeled, not promised. FX is neutral; no extra currency gain enters OP.

Physical and commercial drivers

Quarter	DRAM bit index	DRAM ASP index	Unit-cost index	DRAM GM
Q3 2026	1.0000	1.625195	1.0300	87.01%
Q4 2026	1.0200	1.787715	1.0600	87.85%
Q1 2027	1.0600	1.966486	1.1000	88.54%
Q2 2027	1.1000	2.064810	1.1200	88.88%
Q3 2027	1.1400	2.126755	1.1400	89.02%
Q4 2027	1.1800	2.126755	1.1600	88.82%
Q1 2028	1.2400	2.099657	1.2500	87.80%
Q2 2028	1.3200	2.072904	1.2300	87.84%
Q3 2028	1.4000	2.046492	1.2000	87.98%
Q4 2028	1.4800	2.020417	1.1700	88.13%
Q1 2029	1.5600	2.035403	1.1500	88.42%
Q2 2029	1.6400	2.050499	1.1300	88.71%
Q3 2029	1.7200	2.065708	1.1100	88.99%
Q4 2029	1.8000	2.081029	1.0900	89.27%
Q1 2030	1.8600	2.091357	1.0700	89.52%
Q2 2030	1.9200	2.101737	1.0500	89.76%
Q3 2030	1.9800	2.112167	1.0300	90.01%
Q4 2030	2.0400	2.122650	1.0100	90.25%

Qualification and product allocation

New-fab capacity is a timed, probability-weighted volume component, not a second revenue pool. New/custom products are already within these bits. No unqualified product receives a separate premium ASP increment.

Period		Existing bit index	Potential incremental index		Recognition probability / effective index
2026-27		Visible 1.00 to 1.18	No separate fab increment		Established production; 100% path is still a forecast
2028			1.20	0.05 / 0.15 / 0.25 / 0.35	80%: total 1.24 / 1.32 / 1.40 / 1.48
2029			1.28	0.35 / 0.45 / 0.55 / 0.65	80%: total 1.56 / 1.64 / 1.72 / 1.80
2030			1.40	0.575 / 0.650 / 0.725 / 0.800	80%: total 1.86 / 1.92 / 1.98 / 2.04

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

DRAM

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	134,115.72	87.01%	116,696.87
Q4 2026	150,477.84	87.85%	132,193.12
Q1 2027	172,016.83	88.54%	152,298.01
Q2 2027	187,433.43	88.88%	166,598.45
Q3 2027	200,076.67	89.02%	178,098.47
Q4 2027	207,096.90	88.82%	183,948.43
Q1 2028	214,854.36	87.80%	188,641.53
Q2 2028	225,801.76	87.84%	198,344.24
Q3 2028	236,435.31	87.98%	208,023.98
Q4 2028	246,761.22	88.13%	217,477.26
Q1 2029	262,028.84	88.42%	231,689.60
Q2 2029	277,509.37	88.71%	246,168.97
Q3 2029	293,205.13	88.99%	260,917.68
Q4 2029	309,118.45	89.27%	275,938.07
Q1 2030	321,007.67	89.52%	287,350.39
Q2 2030	333,007.29	89.76%	298,913.69
Q3 2030	345,118.10	90.01%	310,628.78
Q4 2030	357,340.93	90.25%	322,496.46

Other revenue

Quarter	Revenue	Gross margin	Gross profit
Q3 2026		26.23	30.00%7.87
Q4 2026		26.23	30.00%7.87
Q1 2027		27.54	30.00%8.26
Q2 2027		27.54	30.00%8.26
Q3 2027		27.54	30.00%8.26
Q4 2027		27.54	30.00%8.26
Q1 2028		28.85	30.00%8.65
Q2 2028		28.85	30.00%8.65
Q3 2028		28.85	30.00%8.65
Q4 2028		28.85	30.00%8.65
Q1 2029		30.16	30.00%9.05
Q2 2029		30.16	30.00%9.05
Q3 2029		30.16	30.00%9.05
Q4 2029		30.16	30.00%9.05
Q1 2030		31.47	30.00%9.44
Q2 2030		31.47	30.00%9.44
Q3 2030		31.47	30.00%9.44
Q4 2030		31.47	30.00%9.44

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	Model	134,141.95	116,704.74	4,694.97	2,414.56	0.00	109,595.22	81.70%
Q4 2026	Model	150,504.07	132,200.99	5,267.64	2,709.07	0.00	124,224.27	82.54%
Q1 2027	Model	172,044.37	152,306.27	6,021.55	3,096.80	0.00	143,187.92	83.23%
Q2 2027	Model	187,460.97	166,606.72	6,561.13	3,374.30	0.00	156,671.28	83.58%
Q3 2027	Model	200,104.20	178,106.73	7,003.65	3,601.88	0.00	167,501.21	83.71%
Q4 2027	Model	207,124.44	183,956.69	7,249.36	3,728.24	0.00	172,979.10	83.51%
Q1 2028	Model	214,883.21	188,650.18	7,950.68	3,867.90	0.00	176,831.61	82.29%
Q2 2028	Model	225,830.61	198,352.89	8,355.73	4,064.95	0.00	185,932.21	82.33%
Q3 2028	Model	236,464.16	208,032.63	8,749.17	4,256.35	0.00	195,027.10	82.48%
Q4 2028	Model	246,790.07	217,485.91	9,131.23	4,442.22	0.00	203,912.46	82.63%
Q1 2029	Model	262,059.00	231,698.65	9,434.12	4,717.06	0.00	217,547.46	83.01%
Q2 2029	Model	277,539.53	246,178.02	9,991.42	4,995.71	0.00	231,190.88	83.30%
Q3 2029	Model	293,235.29	260,926.73	10,556.47	5,278.24	0.00	245,092.02	83.58%
Q4 2029	Model	309,148.61	275,947.12	11,129.35	5,564.67	0.00	259,253.10	83.86%
Q1 2030	Model	321,039.14	287,359.83	11,236.37	5,778.70	0.00	270,344.76	84.21%
Q2 2030	Model	333,038.76	298,923.13	11,656.36	5,994.70	0.00	281,272.08	84.46%
Q3 2030	Model	345,149.57	310,638.22	12,080.24	6,212.69	0.00	292,345.29	84.70%
Q4 2030	Model	357,372.40	322,505.90	12,508.03	6,432.70	0.00	303,565.16	84.94%

Quarterly drivers and confidence

Calendar / fiscal	Status	Revenue	OP margin	OP	Main driver	Confidence
Q3 2026 / FQ3 FY2026	Model	134,141.95	81.70%	109,595.22	July/Aug actual; September proxy	MEDIUM revenue / LOW margin
Q4 2026 / FQ4 FY2026	Model	150,504.07	82.54%	124,224.27	Contract-price carry-through	MEDIUM revenue / LOW margin
Q1 2027 / FQ1 FY2027	Model	172,044.37	83.23%	143,187.92	Current-fab bits and renewal	MEDIUM revenue / LOW margin
Q2 2027 / FQ2 FY2027	Model	187,460.97	83.58%	156,671.28	Technology conversion	MEDIUM revenue / LOW margin
Q3 2027 / FQ3 FY2027	Model	200,104.20	83.71%	167,501.21	Qualified DRAM volume	MEDIUM revenue / LOW margin
Q4 2027 / FQ4 FY2027	Model	207,124.44	83.51%	172,979.10	Cost discipline	MEDIUM revenue / LOW margin
Q1 2028 / FQ1 FY2028	Model	214,883.21	82.29%	176,831.61	New-fab weighted volume; price normalization	LOW
Q2 2028 / FQ2 FY2028	Model	225,830.61	82.33%	185,932.21	New-fab weighted volume; price normalization	LOW
Q3 2028 / FQ3 FY2028	Model	236,464.16	82.48%	195,027.10	New-fab weighted volume; price normalization	LOW
Q4 2028 / FQ4 FY2028	Model	246,790.07	82.63%	203,912.46	New-fab weighted volume; price normalization	LOW
Q1 2029 / FQ1 FY2029	Model	262,059.00	83.01%	217,547.46	Conditional bit growth and yield	LOW
Q2 2029 / FQ2 FY2029	Model	277,539.53	83.30%	231,190.88	Conditional bit growth and yield	LOW
Q3 2029 / FQ3 FY2029	Model	293,235.29	83.58%	245,092.02	Conditional bit growth and yield	LOW
Q4 2029 / FQ4 FY2029	Model	309,148.61	83.86%	259,253.10	Conditional bit growth and yield	LOW
Q1 2030 / FQ1 FY2030	Model	321,039.14	84.21%	270,344.76	Conditional bit growth and yield	LOW
Q2 2030 / FQ2 FY2030	Model	333,038.76	84.46%	281,272.08	Conditional bit growth and yield	LOW
Q3 2030 / FQ3 FY2030	Model	345,149.57	84.70%	292,345.29	Conditional bit growth and yield	LOW
Q4 2030 / FQ4 FY2030	Model	357,372.40	84.94%	303,565.16	Conditional bit growth and yield	LOW

Annual trajectory cross-check

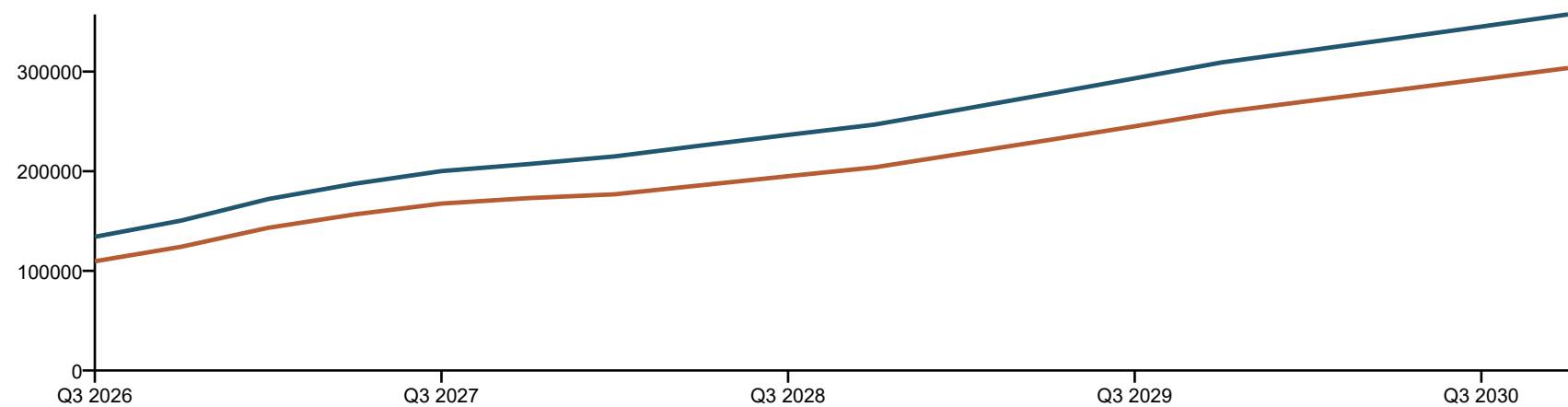
The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin
2027	766,733.98	640,339.51	83.52%
2028	923,968.05	761,703.38	82.44%
2029	1,141,982.44	953,083.47	83.46%
2030	1,356,599.87	1,147,527.29	84.59%

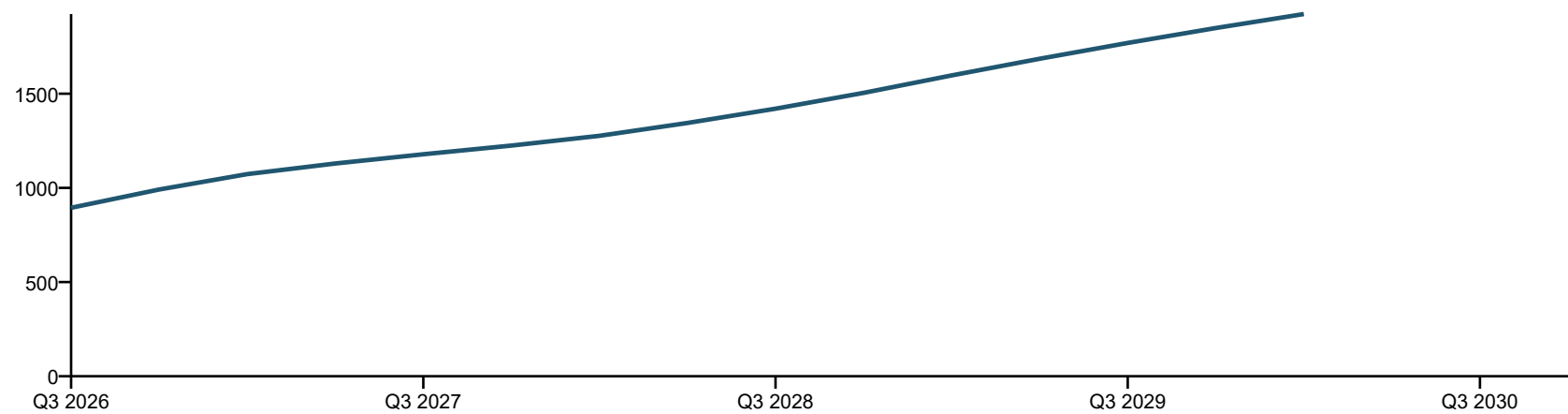
2027 growth is dominated by annualizing the July 2026 price reset, not a sudden doubling of wafer output. The future bit path is deliberately below a simplistic conversion of all announced new wafer starts into instant good-die sales. Later unit-cost increases reflect the new-fab burden. The model generates unusually high gross and operating margins because reported Q2 gross margin was already 79.5% before the July revenue step. That arithmetic is conditional on price retention, not proof of permanently superior economics. A 10% ASP reduction at unchanged volume and manufacturing cost lowers revenue and GP almost dollar-for-dollar before modeled variable opex; this is the main falsification risk. A USD2T all-memory TAM includes NAND and HBM and is not Nanya addressable revenue.

5. Forward four-quarter ceiling and derivations

Revenue (blue), operating profit (orange) - TWD M



Forward-four-quarter ceiling - TWD/share



6x forward-four-quarter operating profit is the user-selected multiple for Nanya as of September 21, 2026, held constant across all windows. Each one-turn change moves ceilings by 16.67%. No multiple expansion is used to amplify the capacity ramp. The existing IFRS-style operating-profit basis and diluted-share denominator are unchanged.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	533,678.70	6x	3,580.631	894.28
Q4 2026	Q4 2026 to Q3 2027	591,584.69	6x	3,580.631	991.31
Q1 2027	Q1 2027 to Q4 2027	640,339.51	6x	3,580.631	1,073.01
Q2 2027	Q2 2027 to Q1 2028	673,983.20	6x	3,580.631	1,129.38
Q3 2027	Q3 2027 to Q2 2028	703,244.13	6x	3,580.631	1,178.41
Q4 2027	Q4 2027 to Q3 2028	730,770.02	6x	3,580.631	1,224.54
Q1 2028	Q1 2028 to Q4 2028	761,703.38	6x	3,580.631	1,276.37
Q2 2028	Q2 2028 to Q1 2029	802,419.24	6x	3,580.631	1,344.60
Q3 2028	Q3 2028 to Q2 2029	847,677.91	6x	3,580.631	1,420.44
Q4 2028	Q4 2028 to Q3 2029	897,742.83	6x	3,580.631	1,504.33
Q1 2029	Q1 2029 to Q4 2029	953,083.47	6x	3,580.631	1,597.07
Q2 2029	Q2 2029 to Q1 2030	1,005,880.76	6x	3,580.631	1,685.54
Q3 2029	Q3 2029 to Q2 2030	1,055,961.95	6x	3,580.631	1,769.46
Q4 2029	Q4 2029 to Q3 2030	1,103,215.22	6x	3,580.631	1,848.64
Q1 2030	Q1 2030 to Q4 2030	1,147,527.29	6x	3,580.631	1,922.89

Recognition timing by ceiling

Ceiling quarter	Key inflection entering the forward window
Q3 2026	July/Aug actual; September proxy through Technology conversion
Q4 2026	Contract-price carry-through through Qualified DRAM volume
Q1 2027	Current-fab bits and renewal through Cost discipline
Q2 2027	Technology conversion through New-fab weighted volume; price normalization
Q3 2027	Qualified DRAM volume through New-fab weighted volume; price normalization
Q4 2027	Cost discipline through New-fab weighted volume; price normalization
Q1 2028	New-fab weighted volume; price normalization through New-fab weighted volume; price normalization
Q2 2028	New-fab weighted volume; price normalization through Conditional bit growth and yield
Q3 2028	New-fab weighted volume; price normalization through Conditional bit growth and yield
Q4 2028	New-fab weighted volume; price normalization through Conditional bit growth and yield
Q1 2029	Conditional bit growth and yield through Conditional bit growth and yield
Q2 2029	Conditional bit growth and yield through Conditional bit growth and yield
Q3 2029	Conditional bit growth and yield through Conditional bit growth and yield
Q4 2029	Conditional bit growth and yield through Conditional bit growth and yield
Q1 2030	Conditional bit growth and yield through Conditional bit growth and yield

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(109,595.22 + 124,224.27 + 143,187.92 + 156,671.28) \times 6 / 3,580.631 = \text{TWD } 894.28$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2026: OP window Q4 2026 to Q3 2027: $(124,224.27 + 143,187.92 + 156,671.28 + 167,501.21) \times 6 / 3,580.631 = \text{TWD } 991.31$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2027: OP window Q1 2027 to Q4 2027: $(143,187.92 + 156,671.28 + 167,501.21 + 172,979.10) \times 6 / 3,580.631 = \text{TWD } 1,073.01$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2027: OP window Q2 2027 to Q1 2028: $(156,671.28 + 167,501.21 + 172,979.10 + 176,831.61) \times 6 / 3,580.631 = \text{TWD } 1,129.38$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2027: OP window Q3 2027 to Q2 2028: $(167,501.21 + 172,979.10 + 176,831.61 + 185,932.21) \times 6 / 3,580.631 = \text{TWD } 1,178.41$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2027: OP window Q4 2027 to Q3 2028: $(172,979.10 + 176,831.61 + 185,932.21 + 195,027.10) \times 6 / 3,580.631 = \text{TWD } 1,224.54$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2028: OP window Q1 2028 to Q4 2028: $(176,831.61 + 185,932.21 + 195,027.10 + 203,912.46) \times 6 / 3,580.631 = \text{TWD } 1,276.37$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2028: OP window Q2 2028 to Q1 2029: $(185,932.21 + 195,027.10 + 203,912.46 + 217,547.46) \times 6 / 3,580.631 = \text{TWD } 1,344.60$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2028: OP window Q3 2028 to Q2 2029: $(195,027.10 + 203,912.46 + 217,547.46 + 231,190.88) \times 6 / 3,580.631 = \text{TWD } 1,420.44$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2028: OP window Q4 2028 to Q3 2029: $(203,912.46 + 217,547.46 + 231,190.88 + 245,092.02) \times 6 / 3,580.631 = \text{TWD } 1,504.33$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2029: OP window Q1 2029 to Q4 2029: $(217,547.46 + 231,190.88 + 245,092.02 + 259,253.10) \times 6 / 3,580.631 = \text{TWD } 1,597.07$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2029: OP window Q2 2029 to Q1 2030: $(231,190.88 + 245,092.02 + 259,253.10 + 270,344.76) \times 6 / 3,580.631 = \text{TWD } 1,685.54$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2029: OP window Q3 2029 to Q2 2030: $(245,092.02 + 259,253.10 + 270,344.76 + 281,272.08) \times 6 / 3,580.631 = \text{TWD } 1,769.46$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2029: OP window Q4 2029 to Q3 2030: $(259,253.10 + 270,344.76 + 281,272.08 + 292,345.29) \times 6 / 3,580.631 = \text{TWD } 1,848.64$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2030: OP window Q1 2030 to Q4 2030: $(270,344.76 + 281,272.08 + 292,345.29 + 303,565.16) \times 6 / 3,580.631 = \text{TWD } 1,922.89$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

Forward diluted denominator = period-end basic 3,450.2059M + Q2 dilutive employee options 121.276M + employee compensation 9.149M = 3,580.6309M. The reported Q2 weighted-average diluted 3,553.586M includes less than a full quarter of the April placement; using it unchanged would understate the current capital base. Flat future dilution is a modeling convention, not a repurchase promise. [N1]

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	1,276.37
5% more shares	1,215.59
10% more shares	1,160.34

6. Cross-book relationships and monitoring

Nanya <-> MU/Samsung/SK hynix: HBM/DDR5 capacity allocation supports mature DRAM pricing; restored legacy output is a counter-signal.

Nanya <-> Winbond: shared specialty DRAM scarcity, different flash/custom mix. Compare company realized prices rather than applying one NAND index to both.

Nanya <-> SNDK/Kioxia: eSSD expansion can consume Nanya controller DRAM; the NAND chips and DRAM chips are distinct revenue pools.

Information processing: monthly data changes the price/recognition model; customer equity improves funding but causes dilution; no other portfolio Base is changed by this deep dive.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Competition	-1	CXMT and incumbent production can normalize ASPs; qualification does not assure permanent scarcity.
Execution	-1	Process yields, commissioning, packaging and customer qualification can delay new bits.
Cycle / demand concentration	-1	A DRAM-dominated earnings stream is sensitive to contract renewal and downstream inventory.
Capital commitments / dilution	-1	Large fab spending and future equity compensation can lower cash returns and per-share earnings.

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
1. September/October 2026 revenue	September near 45,584 model and sustained contract reset	If below the monthly trajectory, replace Q3 revenue with reported data; do not retain stale ASP estimates.
2. Q3 2026 reporting	Gross profit converts the revenue step; no new inventory build	Lower realized margin or rising cost/bit replaces the current cost path.
3. 2027 tool installation / qualifications	Fab and 1C/customer milestones arrive on time	Delay shifts the incremental bit path; reduce 80% weight if a milestone fails.
4. 2028 production	Qualified phase-one output rises toward announced capacity	Only accepted bits enter revenue. Do not equate installed wafers with sold bits.
5. Each competitor/customer report	Tight supply and stable order commitments	Legacy supply restoration, customer destocking or renewals below assumed ASP cuts earnings immediately.

Full Q&A transcripts, customer-level capex purchases and DDR-generation gross margins were unavailable. No verified Yahoo HIGH-estimate series is used as a floor. Near-term revenue has two reported monthly anchors; future volume, mix, cost and price are conditional Plan-B estimates. Precision in the tables is arithmetic. The next 12-16 months carry the principal decision weight; 2028-30 are LOW-confidence extensions.

8. Sources and snapshot

- **N1** - August 2026. Nanya Q2 2026 reviewed consolidated financial statements. Actual revenue, gross profit, expenses and share dilution; values in TWD thousands converted to millions.
- **N2** - July 10, 2026. Nanya Q2 management presentation. Bit/price direction, supply, capex, fab timing and AI application mix; preliminary financials superseded by N1.
- **N3** - April 13, 2026. Nanya Q1 management presentation. Prior-quarter operating context, customer investment and technology roadmap.
- **N4** - September 2, 2026. August and July company monthly revenue. Unaudited monthly actuals; September is a model assumption.
- **N5** - August 2026. July company revenue release. Contract renewal timing and July step in revenue.
- **N6** - April 23, 2026. Nanya 2025 Annual Report, primary filing reproduced by FinancialFilings. Market Status, competitive threats and new-fab construction; historical 2025 share is not a current estimate.
- **N7** - September 21, 2026. Nanya dated market quote. TWD516 at September 21 14:30, header only.
- **U1** - Provided in this conversation. User-provided BofA memory outlook and TrendForce NAND outlook. BofA image: DRAM pricing softens about 5% in 2028. NAND +23%/+20%/+15% is not a Nanya DRAM price schedule.
- **U2** - September 21, 2026. User valuation instruction. Use 6x operating profit for Nanya and Winbond; existing Base scores unchanged.

Model snapshot: September 21, 2026. 2408.TW. Framework 6.0.2. Base 45/60 (75%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/2408TW_NanyaTechnology_DeepDive.pdf and .md. The companion company worksheet uses this same model and links its ceiling series to the master portfolio.